

Quiz 1: KKT Regimes in a Quantity-Quality Household

ECON 803 - Microeconomic Theory I - Fall 2026

October 2, 2026 - 25 minutes - 20 points

Name: Student number:

Show the KKT conditions and enough algebra to identify every active set. The model and all notation needed for the quiz are defined below; no prior knowledge of Galor is required.

A static household quantity-quality problem

Consider a household with 20 units of income. It allocates resources among its own consumption, the number of children, and investment in the children's human capital. More children increase family size (quantity), while more investment per child increases child quality. Because both uses are costly, the household faces a quantity-quality trade-off. Galor studies this mechanism inside a dynamic growth model; here we use a deliberately simple static problem to isolate the KKT economics.

The choice variables are: $c \geq 0$ is parental consumption (the numeraire, with price one); $n \geq 0$ is a divisible fertility index; and $H \geq 0$ is total human-capital investment across all children, not investment per child. When $n > 0$, average quality per child is $q = H/n$. Thus $n = 1/4$ means a near-zero positive value of this index, not literally one-quarter of a child.

The parameters are: $p \geq 0$, the direct resource cost of each unit of fertility; $k \geq 0$, the linear price of total human-capital investment; $s \geq 0$, the minimum required quality per child; and $\underline{n}_r \geq 0$, the fertility floor in regime r . The household chooses $c, H \geq 0$ and $n \geq \underline{n}_r$ to maximize

$$U(c, n, H) = c + 6n - \frac{1}{2}n^2 + 4H - \frac{1}{2}H^2$$

The terms $6n - n^2/2$ and $4H - H^2/2$ give diminishing marginal benefits from family size and total child human capital. The quadratic specification is chosen for tractability, not as a literal description of preferences.

The constraints are

$$c + pn + kH + \frac{1}{2}H^2 \leq 20, \quad sn \leq H \leq 4n, \quad n \geq \underline{n}_r.$$

Budget: income is normalized to 20. Expenditure is parental consumption c , direct child cost pn , linear education cost kH , and the convex cost $H^2/2$. The last term makes additional aggregate investment progressively more expensive. Since more c always raises utility, no income is discarded at an optimum.

Quality cone: $sn \leq H \leq 4n$ links total investment to the number of children. If $n > 0$, division by n gives $s \leq q = H/n \leq 4$. Thus every child must receive at least quality s , while 4 is a normalized upper bound on quality per child. If $n = 0$, the two inequalities force $H = 0$, so the household cannot purchase child human capital without children.

Fertility floor: $n \geq \underline{n}_r$ is zero in the first two regimes. In the last regime it is $1/4$, which produces a positive but near-zero KKT corner.

1. General KKT system [8 points]

- (a) Use multipliers $\lambda, \mu_c, \mu_n, \alpha, \beta \geq 0$ for, respectively, the budget slack, $c \geq 0$, $n - \underline{n}_r \geq 0$, $H - sn \geq 0$, and $4n - H \geq 0$. Write the Lagrangian and the complete KKT system. [6]
- (b) Explain why these conditions are sufficient. Show that whenever $c^* > 0$, the budget binds and $\lambda^* = 1$. [2]

2. Solve three regimes [9 points]

For each regime, find (c^*, n^*, H^*) , report $q^* = H^*/n^*$ when it is defined, identify the active inequalities, and give one valid multiplier vector. You may first use the binding budget to eliminate consumption. [3 points each]

Quantity-dominant (Malthusian): $(p, k, s, \underline{n}_r) = (2, 5, 0, 0)$.

Modern quantity-quality balance: $(p, k, s, \underline{n}_r) = (4, 2, 0, 0)$.

High-quality-cost, low-fertility (post-modern) extension: $(p, k, s, \underline{n}_r) = (4, 6, 2, 1/4)$.

3. Interpretation [3 points]

- (a) Describe the movement from quantity to quality across your first two solutions. [1]
- (b) Explain why a high quality price k alone need not eliminate fertility, but a high k together with the minimum-quality requirement $H \geq 2n$ can push fertility down to its small lower bound. [1]
- (c) A full growth model would let today's household choices affect tomorrow. Name one plausible feedback---for example, education could raise the next generation's human capital, fertility could change future population, or technology could change the return to education. Then explain why the binding low-fertility floor is our teaching extension rather than Galor's prediction. [1]